🏨 Hostel Room Allocation System

An AI-inspired desktop application that automates hostel room allocation using Constraint Satisfaction Problem (CSP) principles and Preference Matching.

Python

Tkinter

SQLite

Status

License

📌 Overview

The **Hostel Room Allocation System** is a desktop-based application developed during my **M.Tech (AI) Internship**. The project automates the hostel room allocation process by intelligently assigning students to rooms while satisfying essential constraints and maximizing roommate compatibility.

Instead of relying on manual allocation, the system applies **Constraint Satisfaction Problem (CSP)** concepts along with a **Preference Matching Algorithm** to produce fair, efficient, and practical room assignments.

🎯 Problem Statement

Traditional hostel room allocation is often performed manually, making the process:

- Time-consuming
- Error-prone
- Difficult to manage for large numbers of students
- Unable to consider multiple student preferences simultaneously
- Inefficient when accessibility requirements must be satisfied

This project addresses these challenges through intelligent automation.

💡 Solution

The system evaluates both **hard constraints** and **soft constraints** before assigning rooms.

Hard Constraints

- Room capacity
- Gender compatibility
- Accessibility requirements
- Room availability

Soft Constraints

- Lifestyle preferences
- Study habits
- Sleep schedule compatibility
- Roommate preference score

Each allocation receives a **compatibility score**, helping ensure students are placed in the most suitable room available.

✨ Features

- 👨‍🎓 Student Management
- Add Student
- Update Student
- Delete Student

- View Student Records

- 🏠 Room Management

- Add Rooms
- Edit Rooms
- Delete Rooms

- Track Availability

- 🤖 Intelligent Allocation

- Automatic room assignment
- Constraint Satisfaction logic
- Preference Matching Algorithm

- Compatibility scoring

- ♿ Accessibility Support

- Prioritizes students with accessibility requirements

- Ensures appropriate room allocation

- 📊 Database Integration

- Persistent data storage using SQLite

- Automatic record management

- 🎨 User-Friendly Interface

- Clean Tkinter GUI
- Responsive forms
- Real-time feedback
- Easy navigation

🧠 AI Concepts Used

Although this is an internship-level desktop application, it demonstrates practical implementation of several Artificial Intelligence concepts:

- Constraint Satisfaction Problem (CSP)
- Rule-Based Decision Making
- Preference Matching
- Optimization under Constraints
- Compatibility Scoring

🛠️ Technology Stack

Technology	Purpose
Python	Core Programming Language
Tkinter	Desktop GUI
SQLite	Database
OOP	Project Architecture
SQL	Data Storage & Retrieval

📂 Project Structure

```
Hostel-Room-Allocation-System/
│
├── database/
│   ├── hostel.db
│   └── database.py
│
├── models/
│   ├── student.py
│   ├── room.py
│   └── allocation.py
│
├── gui/
│   ├── dashboard.py
│   ├── students.py
│   ├── rooms.py
│   └── allocation.py
│
├── assets/
│
├── main.py
│
├── requirements.txt
│
└── README.md
```

⚙️ Installation

Clone the repository

```
git clone https://github.com/yourusername/Hostel-Room-Allocation-System.git
```

Move into the project directory

```
cd Hostel-Room-Allocation-System
```

Install dependencies

```
pip install -r requirements.txt
```

Run the application

```
python main.py
```

🚀 Future Improvements

- Login & Authentication
- Multi-Hostel Support
- Advanced AI Optimization Algorithms
- Web-Based Version
- Analytics Dashboard
- PDF Report Generation
- Email Notifications
- Cloud Database Integration

📸 Screenshots

Add screenshots of:

- Dashboard
- Student Management
- Room Management
- Allocation Window
- Compatibility Results

🎓 Learning Outcomes

Through this project, I strengthened my understanding of:

- Object-Oriented Programming (OOP)
- Python GUI Development
- SQLite Database Design
- AI-based Constraint Satisfaction Problems
- Preference Matching Algorithms
- Desktop Application Development
- Software Architecture

- Debugging and Testing
- Real-World Problem Solving

👨‍💻 Author

Abdullah Rafique

BS Artificial Intelligence Student

M.Tech AI Intern

Passionate about Artificial Intelligence, Python Development, and building intelligent software solutions that solve real-world problems.

🤝 Acknowledgements

Special thanks to my internship mentors and instructors for their continuous guidance, valuable feedback, and encouragement throughout the development of this project.

⭐ Support

If you found this project useful, consider giving it a ⭐ **Star** on GitHub.

Your support motivates me to build more real-world AI and software development projects.

📄 License

This project is developed for educational and internship purposes.